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Article

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Sulfurization of Dissolved Organic Matter Increases Hg-Sulfide-DOM Bioavailability to a Hg-Methylating Bacterium

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Abstract

Reactions of dissolved organic matter (DOM) with aqueous sulfide (termed sulfurization) in anoxic environments can substantially increase DOM's reduced sulfur functional group content. Sulfurization may affect DOM-trace metal interactions, including complexation and metal-containing particle precipitation, aggregation, and dissolution. Using a diverse suite of DOM samples, we found that susceptibility to additional sulfur incorporation via reaction with aqueous sulfide increased with increasing DOM aromatic-, carbonyl-, and carboxyl-C content. The role of DOM sulfurization in enhancing Hg bioavailability for microbial methylation was evaluated under conditions typical of Hg methylation environments (μM sulfide concentrations and low Hg/DOM molar ratios). Under conditions of predicted metacinnabar supersaturation, microbial Hg methylation increased with increasing DOM sulfurization, likely reflecting either effective inhibition of metacinnabar growth/aggregation and/or formation of Hg(II)-DOM thiol complexes with high bioavailability. Remarkably, Hg methylation efficiencies with the most sulfurized DOM samples were similar (>85% of total Hg methylated) to that observed in the presence of L-cysteine, a ligand facilitating rapid Hg(II) biouptake and methylation. This suggests that complexes of Hg(II) with DOM thiols have similar bioavailability to Hg(II) complexes with low molecular weight thiols. Overall, our results are a demonstration of the importance of DOM sulfurization to trace metal/metalloid (especially mercury) fate in the environment. DOM sulfurization likely represents another link between anthropogenic sulfate enrichment and MeHg production in the environment.

1. Introduction

Dissolved organic matter (DOM) plays a key role in the mobility and bioavailability of mercury (Hg) in aquatic systems. At the ecosystem scale, the flux of mercury in coastal[1] and terrestrial[2] ecosystems is tightly coupled to dissolved organic carbon (DOC) flux, as inorganic Hg(II) ($Hg(II)_i$) forms strong complexes with thiol moieties in DOM.1[3] Within the anoxic, often sulfidic, environments, where Hg methylation occurs, such as bottom sediments, wetlands, and flooded soils, the importance of DOM is likely in slowing the growth and/or aggregation of β-HgS (metacinnabar) particles.[4-7] Inhibition of β-HgS growth/aggregation is hypothesized to support increased rates of microbial Hg methylation,[8-10] as smaller Hg-S clusters/particles are more bioavailable to Hg-methylating bacteria,[10] a process which remains incompletely understood. Additionally, dissolved Hg-DOM complexes may be bioavailable for microbial Hg uptake and subsequent methylation.[11,12]

Hg methylation by a model Hg-methylating bacterium suspended in Hg-sulfide-DOM solutions correlated with both DOM aromaticity and sulfur content.[9] This suggested the importance of both nonspecific (increased inhibition of β-HgS precipitation/aggregation by more aromatic and surface active DOM) and specific (capping of β-HgS by DOM thiols or formation of Hg-DOM thiol complexes) interactions between DOM and Hg in controlling Hg bioavailability under sulfidic conditions. In this paper, we further explore the relationship between DOM composition and Hg bioavailability for methylation. Specifically, we focus on how diagenetic sulfurization of DOM (the incorporation of sulfur into DOM) influences the Hg-sulfide-DOM interactions that determine, in part, the rate and extent of Hg methylation in anoxic environments.

In a broader context, most studies on the role of DOM in trace metal complexation and metal-containing particle growth and aggregation have focused on DOM isolated from oxic surface waters. There is growing recognition that DOM isolated from oxic surface waters may not be representative of that found in highly stratified lakes, sediment porewaters, and groundwaters where sulfate reduction is an important biogeochemical process. Sulfide produced from microbial sulfate reduction can be readily incorporated into DOM via S_N2, S_NAr, and Michael addition reactions, as demonstrated in a number of studies in which sulfide was reacted with model compounds[13,14] as well as extracted natural DOM samples.[15-17] High resolution mass spectrometric studies of DOM composition in the North American prairie pothole region[18] and the Florida Everglades[19,20] indicate that CHOS and CHONS compounds can reach 10 to nearly 50% of all identified molecular formulae. Sulfurization is relatively rapid, occurring on a timescale of days for environmentally relevant concentrations of DOM and sulfide.[17] The products of sulfurization identified by X-ray absorption near edge spectroscopy (XANES) include both reduced (thiol and disulfide species) as well as oxidized (e.g., sulfoxides, sulfones, sulfonates, and sulfate esters) species, though there is evidence that the proportion of reduced S species increases with increasing degree of sulfurization.[16,20]

Here we describe experiments where the sulfur content of a suite of DOM samples was directly manipulated by reacting DOM samples with hydrogen sulfide/bisulfide at room temperature. We subsequently evaluated microbial Hg-methylation in solutions containing mg/L concentrations of our suite of the sulfurized DOM samples and μM concentrations of sulfide, mimicking conditions found in many anoxic environments. Our experiments provide information on the role of DOM sulfurization in controlling Hg bioavailability to Hg-

methylating bacteria and shed additional light on the mechanisms of $Hg(II)_i$ uptake and methylation in sulfidic environments.

2. Experimental Section

2.1 Dissolved Organic Matter Samples

Dissolved organic matter samples with a range of average molecular weight, aromaticity, and native sulfur content were purchased from the International Humic Substances Society (IHSS) for use in the DOM sulfurization experiments. The four samples included humic and fulvic acids from a blackwater river draining the Okeefenokee Swamp in Georgia (Suwanee River humic and fulvic acids; SRHA standard II and SRFA standard II), a fulvic acid from a high latitude (60 °N) drinking water reservoir in Norway (Nordic Lake fulvic acid; NLFA); and a microbially-derived fulvic acid from a eutrophic Antarctic pond (Pony Lake fulvic acid; PLFA). The PLFA sample is unique amongst our samples in its high native sulfur content (3.03% by mass compared to 0.4-0.6% for the other DOM samples). Sulfur speciation determined by XANES was available for three of the four DOM samples (SRHA, SRFA, and PLFA); reduced exocyclic and heterocyclic sulfur comprised 53-68% of total sulfur for these samples, and is likely similar for NLFA based on analysis of Nordic Lake humic acid (71.8% of total S as reduced S).[21] Characteristics of the DOM samples are summarized in **Table 1.**

2.2 DOM Sulfurization Reactions

Inside an anoxic glove bag (Coy Laboratory Products) with an atmosphere of 2-5% H_2 and 95-98% N_2, maintained O_2-free with Pd-catalysts, DOM samples were dissolved to ~500 mg C/L in N_2-degassed deionized water (DDIW). The stock solution was aliquoted into foil-wrapped borosilicate glass serum bottles and solutions were spiked with Na_2S to give total

sulfide (H_2S_T) additions of 0 to 50 mmol S/mol C, with four to five different S/C ratios evaluated for each DOM sample. The pH of each solution was adjusted to the pH of the unspiked DOM solution (~5.0) with degassed trace metal grade HNO_3. The serum bottles (containing a liquid: headspace ratio of >10:1) were then immediately stoppered and allowed to react for 48 h at room temperature. The pH and range of S/C additions were similar to that employed in a study of As sorption to sulfurized OM.[16] Notably, Hoffmann et al.[16] found minimal impact of pH on DOM sulfurization in the pH range 5.0-7.0. The incubation time (48 h) was selected to yield maximal S incorporation based on kinetic investigations of S incorporation into DOM.[16,17]

Following the 48 h reaction period, sulfurized DOM was recovered by solid phase extraction (SPE)[22] carried out in the anoxic glove bag. Sulfurized DOM samples were acidified to pH 2.0 with degassed 50% v/v trace metal-grade HCl, and then passed through conditioned 3 mL Agilent Bond Elut PPL columns (100 mg of styrene-divinylbenzene polymer; pore size 150 Å) at a flow rate of ~2 mL/min. Salts (including unreacted Na_2S) were removed by washing with two column volumes of 0.01 M degassed HCl. The columns were then dried by drawing an anoxic atmosphere through the columns at constant vacuum. Sulfurized DOM was eluted with 10 mL of HPLC grade methanol (Fisher). Methanolic DOM solutions were evaporated under a gentle N_2-stream and then reconstituted in DDIW inside the glove bag. Similar to previous reports,[22] DOM recovery ranged from 32 to 81% of total DOC among all DOM samples (mean = 53±18%), with lower variability (3-17% RSD) for each sub-treatment of a single DOM sample (recoveries for individual samples reported in Supporting Information **Table SI-1**). Analysis of UV-Vis spectra for sulfurized samples indicated that our sulfurization

and SPE procedure did not substantially alter the size and/or aromaticity of a given DOM isolate. For example, for a given isolate, no correlations were observed between sulfur incorporation and the slope ratio[23] of the DOM sample recovered by SPE (reported in Table SI-1 of Supporting Information). Sulfurized DOM samples were stored in airtight stoppered bottles at 4 °C for no more than five days prior to initiation of Hg bioavailability assays. For each set of DOM samples, we included an unsulfurized DOM control subjected to the identical SPE recovery procedure as the sulfurized samples. SPE recovery and optical properties of the unsulfurized DOM samples were quite similar to that of sulfurized samples.

2.3 Hg Methylation Assays

Hg bioavailability for microbial methylation was determined in washed-cell assays in polypropylene tubes with the model organism *Desulfovibrio desulfuricans* ND132, an efficient Hg-methylator[24,25] previously used to study Hg methylation in solutions containing DOM and sulfide.[8,9] Strain ND132 was grown to mid-log phase on an estuarine pyruvate-fumarate (EPF) medium[24] at 31 °C, centrifuged at 3000 *g* for 15 min to pelletize cells, then resuspended in a minimal pyruvate-fumarate media (composition in Graham et al.[8]). Cells were centrifuged a second time and methylation assays were initiated by resuspending cells in filter-sterilized minimal media that had been amended with 0.41 nM enriched inorganic $^{201}Hg(II)_i$ (in 1% v/v HCl) and ~10 mg C/L of sulfurized DOM, reduced with 25 μM titanium nitrilotriacetic acid (TiNTA) to remove trace oxygen contamination, and pre-equilibrated for 24 h at 31 °C. The 24 h pre-equilibration period was selected based on kinetic investigations of Hg-DOM complexation.[26] No external additions of sulfide were necessary, as strain ND132 cleaves the S-C bond in cysteine provided in growth medium releasing low (and reproducible) μM sulfide

concentrations (range of 0.1 to 5.9 μM) in the cysteine-free (excepting positive controls described below) minimal assay medium. As noted below, a portion of the $^{201}Hg(II)_i$ spike was lost to bottle wall adsorption during the ^{201}Hg/DOM pre-equilibration period, and observed total ^{201}Hg concentrations were typically between 0.1 to 0.3 nM. Cell suspensions were sampled immediately for measurement of pH, optical density at 660 nm (OD_{660}), and sulfide (H_2S_T), and then placed in a 31 °C incubator inside the glove bag for 3 h. At the end of the incubation period, cell suspensions were sampled for total Hg (THg), total MeHg, filter-passing (0.2 μm nylon membrane) THg, pH, OD_{660}, and H_2S_T. Three-hour experiments were selected based on previous kinetic experiments demonstrating a plateau in MeHg production by strain ND132 within this time frame.[25] Control experiments included DOM-free controls, positive controls with 500 μM L-cysteine, conditions that favor high rates of MeHg production[25,27], and unsulfurized-DOM controls. Our previous work demonstrated that Hg methylation was insignificant (<0.2% of ^{201}Hg spike methylated) in abiotic controls including DOM,[8] and these controls were not repeated for this investigation. All experiments, including controls, were performed in triplicate except for experiments with the NLFA isolate, which were performed in duplicate.

2.4 Analytical Methods

Total S content in sulfurized DOM samples was determined by inductively coupled plasma mass spectrometry (ICP-MS; Agilent 7500 CE), with monitoring of S at $m/z = 34^+$ and calibration against Na_2SO_4 standards following blank correction. While an isobaric interference at $m/z = 34^+$ due to $^{16}O^{18}O$ limits trace S determination by ICP-MS without collision/reaction cell technology, S concentrations in our DOM samples could readily be

determined by monitoring at *m/z* = 34^+ following blank correction, as sample concentrations (60-470 µM) were substantially above the method detection limit (19.6 µM, based on three times blank standard deviation). Relative percent difference (RPD) for duplicate analysis of S concentration averaged 12.4±14.3% (*n* = 4 pairs). Dissolved organic carbon (DOC) concentrations of DOM samples recovered by SPE (along with initial stocks prior to sulfurization and SPE) were determined as non-purgeable organic carbon using a Shimadzu TOC-V analyzer. Experimentally determined S/C ratios for DOM samples were calculated based on measured S and DOC concentrations.

Unfiltered cell suspensions were analyzed for THg by first digesting 1 mL of sample with 5 mL of 7:4 HNO_3:H_2SO_4 at ~190 °C and then amending with 1% v/v BrCl. Filtered samples were digested overnight in 1% BrCl at room temperature. THg analysis was carried out using on-line reduction with $SnCl_2$ and introduction of elemental Hg vapor directly into the ICP-MS.[8] MeHg concentrations were determined by steam distillation, aqueous phase ethylation with sodium tetraethylborate, and gas chromatography (GC)-ICP-MS using isotopically-enriched $Me^{199}Hg$ as an internal standard.[28] Herein, we report concentrations of excess ^{201}THg and $Me^{201}Hg$ after correction for contributions of 1) ambient THg or MeHg and 2) internal standard $Me^{199}Hg$. We routinely analyzed duplicate samples, standard reference materials (SRMs), and blanks as part of our quality assurance/quality control (QA/QC) efforts; a summary of QA/QC data is presented in Supporting Information **Table SI-2**.

Sulfide was analyzed using a silver/sulfide ion specific electrode with a Ag/AgCl reference electrode (Thermo Scientific) and was calibrated against Pb-titrated Na_2S standards made in sulfide antioxidant buffer (SAOB).[29] Cell densities were estimated based on

193 measurement of OD_{660} in 1 cm cuvets and calibration between OD_{660} and cell counts
194 determined using a Coulter counter (Beckman Coulter Multisizer 4).[9]
195
196 *2.5 Equilibrium Speciation Modeling*
197 All equilibrium speciation modeling was performed in MINEQL+ v. 4.5 (Environmental
198 Research Software). A detailed description of the assumptions and thermodynamic data used
199 in the equilibrium speciation modeling are provided in the Supporting Information. Briefly,
200 equilibrium constants for complexation of inorganic Hg(II) ($Hg(II)_i$) with sulfide were taken
201 from Drott et al.,[30] including a revised value of the equilibrium constant for β-HgS(s)
202 (metacinnabar) solubility. The equilibrium constant for $Hg(II)_i$ complexation by DOM thiols
203 was taken from Skyllberg.[3] Thiol concentrations were estimated based on measured S/C ratio
204 in DOM samples and the percentage of total S as reduced exocyclic S. All other equilibrium
205 constants were taken directly from the MINEQL+ database.

206 ## 3. Results and Discussion

207 *3.1 Sulfur Incorporation into DOM*
208 Reaction of Na_2S with DOM resulted in DOM with appreciably greater total S content
209 (**Figure 1a**), although the yield of sulfurized DOM varied amongst the different DOM isolates.
210 For SRHA and SRFA samples, S content increased from ~2-3 mmol S/mol C for unsulfurized
211 DOM (slightly lower than IHSS's report of native S content of 4.14 and 3.15 mmol S/mol C for
212 SRHA and SRFA, respectively; see Table 1) to ~6 mmol S/mol C at the highest added H_2S_T/DOC
213 ratios. At the highest added H_2S_T/DOC ratios (40-50 mmol H_2S_T/mol C), less than 10% of
214 added H_2S is incorporated into SRHA or SRFA, and sulfur incorporation plateaus at an addition

of 20-30 mmol S/mol C. For the NLFA sample, total S content increased from 3.1 mmol S/mol C for unsulfurized DOM (3.3 mmol S/mol C reported by IHSS) to 12.8 mmol S/mol C. It is unclear whether the observed maximum S content of 12.8 mmol S/mol C represents the limit of total S incorporation for the NLFA sample, as a plateau in S incorporation was not obvious for this sample. Lastly, the PLFA sample, which had the highest native S content (measured as 11.0 mmol S/mol C; IHSS value = 21.7 mmol S/mol C), showed the lowest degree of S incorporation upon sulfide addition. While we obtained good agreement for total native S content for other samples with the reported IHSS elemental composition, our measure for native S content of PLFA was about 50% lower than the reported value. The proportion of S-containing molecules could be lower in our SPE-recovered DOM fraction compared to the original IHSS-isolated material. However, we have attempted to control for fractionation effects by comparing sulfurized DOM samples to unsulfurized samples having undergone the same SPE extraction and recovery procedure. We also note that for a given DOM sample, there was no relationship between S content and DOM recovery or DOM slope ratio (Supporting Information Table SI-1), suggesting that we extracted a similar DOM molecular fraction regardless of degree of sulfurization. The low extent of S incorporation for the PLFA sample may reflect the fact that this sample was already highly sulfurized prior to sulfide addition, and the finite sulfur incorporation capacity of the DOM had already been nearly reached. Sulfate reduction (as evidenced by H_2S odor and microbial species identified by 16S rRNA) which should enhance the sulfurization of DOM is known to occur in Pony Lake.[31]

While there are clear differences in sulfur incorporation capacity amongst DOM samples, the chemical bases of these differences are not well understood. Sulfurized OM can

be produced by reactions of sulfide with ketones,[14] aldehydes,[14] carbohydrates,[32], quinones[13], and other compounds containing one or more unsaturated carbon centers. For DOM, even when high-resolution MS data are available, we lack information regarding the structure and concentration of individual compounds that contribute to the DOM pool, and must rely on bulk measures of DOM quality. One such measure is functional group content derived from ^{13}C NMR. Shown in **Figure 1b**, is the relationship between the sum of the percentage of carbonyl, carboxyl, and aromatic C and the S incorporation capacity. Here, S incorporation capacity is defined as the maximum observed S content (taken as the mean of the plateau in S content if evident; Figure 1a) following reaction with an excess of sulfide minus the native S content. Included in Figure 1b are data from Hoffmann *et al.*[16] and Poulin et al.[20] for which both S incorporation capacity of DOM and ^{13}C NMR data were reported or could be estimated. The data of Poulin *et al.*[20] are noteworthy for representing a "natural experiment" where S content of porewater DOM was measured along a natural porewater sulfide gradient in the Florida Everglades. The S incorporation capacities observed for the DOM samples employed in this study are consistent with that reported previously.[16,20] More interestingly, the proportion of carbonyl, carboxyl, and aromatic functional groups is a reasonably good predictor of susceptibility to abiotic S incorporation. The linear relationship between Σ % carbonyl, carboxyl, and aromatic C and S incorporation capacity (r^2 = 0.86, p < 0.01) suggests that DOM quality, along with sulfide concentration, will play a role in the extent of DOM sulfurization during diagenesis. One caution regarding interpretation of Figure 1b is that our dataset is dominated by DOM samples of terrestrial origin. Further work with DOM from autochthonous sources with a lower proportion of aromatic and carbonyl C[33] is needed to

validate whether bulk measures of aromatic, carbonyl, and carboxyl functional group content can be used to accurately predict DOM sulfur incorporation capacity.

Our study did not include determination of S speciation in the sulfurized DOM. However, other work clearly shows that reduced S content of DOM increases approximately linearly with increased total S incorporation.[16,20] We thus have high confidence that total DOM S is a good proxy for reduced S functional groups important to Hg biogeochemistry.

3.2 Impact of DOM Sulfurization on Hg Methylation in Hg-DOM-Sulfide Solutions

As a case study for the importance of DOM sulfurization on trace element speciation and bioreactivity, we evaluated Hg methylation by *D. desulfuricans* ND132 in solutions containing $^{201}Hg(II)_i$, sulfurized DOM, and low levels of sulfide (3.9±2.2 μM across all experiments; see **Table SI-4** in Supporting Information for concentrations in individual experiments). **Figure 2a** shows total Hg concentration in the experiments as a function of the degree of sulfurization (expressed as S/C ratio of DOM) for each DOM sample. Significant (up to 75%) loss of the $^{201}HgCl_2$ spike (dashed line in Figure 2a) is mostly attributed to $^{201}Hg(II)_i$ sorption to polycarbonate sample containers during the pre-equilibration of the $^{201}HgCl_2$ spike with DOM (or no DOM or 500 μM L-cysteine in controls) in the minimal medium. We confirmed this in a subset of experiments (with SRFA and PLFA) by directly measuring ^{201}Hg on bottle walls. This was done by adding 1% v/v BrCl + 1% v/v HCl directly to bottles used for pre-equilibrating $^{201}HgCl_2$ with sulfurized DOM in the minimal medium, in order to desorb ^{201}Hg from bottle walls. Accounting for $^{201}Hg(II)_i$ recovered from bottle walls, we recovered a total of 81.9±9.9% of total added $^{201}HgCl_2$, similar to what we reported in previous studies.[8,9] The sorption is likely hydrophobic partitioning of neutral $Hg(SR)_2$ complexes onto the plastic

bottle surfaces. Our recovery of sorbed Hg indicates that losses of ^{201}Hg due to reduction by DOM[34,35] and evasion of Hg(0) are not significant in these experiments. While the extent of $^{201}Hg(II)_i$ sorption to bottle walls varied amongst DOM samples, DOM sulfurization had minimal impact on the extent of bottle wall sorption, with the exception of the experiment with SRFA, for which we observed lower bottle wall sorption with increasing DOM sulfurization. Thus, the initial character of each DOM isolate had a larger impact on Hg sorption to bottle walls than did the extent of its sulfurization.

For all DOM samples, Hg methylation by strain ND132 in Hg-sulfide solutions was greater with DOM (range of 16.1-89.8% of ^{201}THg methylated) compared to DOM-free Hg-sulfide solutions (5.5±5.3% ^{201}THg methylated; **Figure 2c**). A similar conclusion is reached when comparing cell-normalized MeHg production (**Figure 2d**; 5-80 x 10^{-23} mol/cell in DOM-containing solutions vs. 1.1±0.9 x 10^{-23} mol/cell in DOM-free experiments). A DOM-dependent enhancement in Hg-methylation of 3- to 16-fold under mildly sulfidic conditions ($[H_2S]_T$ = 3.4±1.6 μM) is consistent with previous findings with a chemically diverse suite of DOM isolates evaluated under similar conditions.[9] Interestingly, in some experiments with sulfurized DOM, Hg methylation efficiency (based on % ^{201}THg methylated) approaches, or even exceeds, that observed in positive controls with 500 μM L-cysteine (mean %MeHg of 79.7±27.9%; Figure 2c), a ligand known to facilitate high rates of $Hg(II)_i$ biouptake in strain ND132[25,36] and other bacteria.[27,37] That similar efficiencies of Hg methylation (and, hence, biouptake) can be achieved in Hg-sulfide-DOM solutions highlights the magnitude of the DOM-dependent enhancement in $Hg(II)_i$ bioavailability.

For three out of four DOM samples evaluated (SRHA, SRFA, and NLFA), absolute MeHg production (Figure 2b), %MeHg (^{201}THg as MeHg; Figure 2c), and cell-normalized MeHg production (Figure 2d) increase with increasing sulfurization of DOM. For these three isolates, %MeHg was linearly correlated with the S/C ratio for each DOM sample (r^2 = 0.94 for SRHA, 0.64 for SRFA, and 0.68 for NLFA), and %Hg methylation increased 1.5-3 fold upon additional DOM sulfur incorporation. For PLFA, which exhibited the smallest relative change in sulfur content upon reaction with sulfide (Figure 1), there was no significant effect of sulfurization on MeHg production by strain ND132 in DOM-sulfide solutions. In our previous work, we identified DOM sulfur content as a potentially important variable in controlling $Hg(II)_i$ bioavailability in Hg-sulfide-DOM solutions based on a multiple linear regression analysis of DOM properties vs. Hg methylation.[9] Here, by directly modifying DOM sulfur content via low temperature reaction with sulfide that mimics diagenetic sulfurization of DOM, we directly confirm that DOM sulfur content contributes to the propensity of DOM to enhance Hg methylation in sulfidic solutions. In addition to the earlier study by Hoffmann *et al.* that showed increased arsenite sorption following DOM sulfurization,[16] our study is among the first to demonstrate the importance of DOM sulfurization to trace metal/metalloid fate.

3.3 Mechanisms of DOM-enhanced Hg methylation

DOM may enhance microbial Hg methylation by multiple causal mechanisms, including: 1) stimulation of microbial metabolism; 2) increasing cell membrane permeability to other solutes;[38] 3) inhibition of growth/aggregation of HgS(s) (metacinnabar),[4-7] with increased bioavailability of nanoscale HgS(s) relative to bulk HgS(s);[10,39,40] and 4) formation of specific $Hg(II)_i$-ligand complexes (e.g., $Hg(II)_i$-thiol complexes) with high bioavailability for

uptake and subsequent methylation.[12] While all of these mechanisms may be operative concurrently, experimental evidence[8] suggests that stimulation of microbial metabolism and increased cell wall/membrane permeability are unimportant in these short-term experiments. Furthermore, DOM sulfurization is likely to increase DOM recalcitrance to microbial utilization (thus contributing to organic matter preservation in natural environments),[41] and it is unlikely that sulfurization would alter the surfactant-like properties of DOM critical to DOM accumulation at cell surfaces.[38] Thus, inhibition of HgS(s) growth/aggregation and/or high bioavailability of Hg-DOM thiol complexes are the most plausible explanations for increased Hg methylation in the presence of sulfurized DOM. We discuss each of these possibilities below.

Earlier work on metal sulfide dissolution[33] and precipitation/aggregation[4,42] emphasized the importance of non-specific, steric interactions related to DOM aromaticity/molecular weight in controlling DOM's inhibition of metal sulfide precipitation/dissolution or enhancement of dissolution, but these studies were performed at high metal/DOM ratios where contributions of low abundance thiol moieties with well-documented impacts on metal sulfide growth/aggregation[5,43,44] may have been less evident. Using the most recent thermodynamic data for metacinnabar solubility,[30] Hg-sulfide[30] and Hg-DOM thiol complexation[3] (summarized in Supporting Information **Table SI-3**), we predict metacinnabar precipitation in all experiments with SRHA and NLFA, some of the SRFA experiments, and none of the PLFA experiments, where measured sulfide concentrations were greatest (**Table SI-5** in Supporting Information). The solubility product for metacinnabar is a source of considerable uncertainty in this model,[45] and the log K_{sp} = 36.8 recommended

by Drott *et al.*[30] is toward the lower end of the uncertainty range for log K_{sp} reported in the NIST Critical database (log K_{sp} = 38.0±2.0).[46] A major obstacle to reliable measurement of β-HgS(s) solubility is distinguishing nanoparticulate Hg from truly dissolved Hg – Drott et al.[30] used 20 nm pore size filters in their determination of β-HgS(s) solubilty, but primary β-HgS(s) particles can be as small as 2-5 nm in diameter. [7,47] Inclusion of nanoparticulate β-HgS(s) in the dissolved fraction can lead to overestimates of β-HgS(s) solubility. Highlighting the sensitivity of the model results to the value of the metacinnabar solubility product, an increase in log K_{sp} from 36.8 to 38.0 results in predicted metacinnabar supersaturation across all experimental treatments. Given that Hg methylation increased with increasing DOM sulfurization under conditions of metacinnabar supersaturation, we posit that sulfurization has the same effect as increasing DOM concentration[8] on $Hg(II)_i$ bioavailability and that Hg methylation efficiency increases with increasing DOM thiol/Hg molar ratio. Shown in the Supporting Information (**Figure SI-1**) are methylation efficiencies for sulfurized SRHA superimposed upon data for SRHA at various DOM/THg ratios from Graham et al.[8] When the DOM concentration is expressed as DOM thiol concentration (based on measured S/C ratio and the assumption that 23.6% of SRHA sulfur is as exocyclic S species), all of the SRHA data collapse onto a single line (slope = 0.57, r^2 = 0.89, p < 0.001). In our previous evaluation of Hg methylation in Hg-sulfide-DOM solutions, we noted that DOM S content was an important predictor of DOM's ability to enhance Hg methylation, and hypothesized that DOM thiols might act as capping agents preventing metacinnabar growth/aggregation.[9] Our direct manipulation of DOM thiol content confirms the important role of DOM thiols even under conditions where metacinnabar formation is likely, and where Hg-DOM thiol complexes are a

small fraction (<0.01 to 2.4%; Supporting Information Table **SI-5**) of total $Hg(II)_i$. The exact mechanism of enhanced bioavailability of nanoscale HgS relative to bulk forms is unknown, but an increased rate of HgS dissolution/ligand exchange at the cell surface for nanoscale HgS capped by DOM thiols is one possibility. Biochemical pathways of $Hg(II)_i$ uptake by Hg-methylating bacteria are poorly understood, though essential trace metal transporters (e.g. for Zn) may be involved.[48]

An alternative hypothesis is that sulfurization of DOM increases the pool of $Hg(II)_i$-ligand complexes that are preferentially taken up by strain ND132. Past workers suggested that $Hg(II)_i$ bioavailability for methylation could be predicted by the concentration of neutral $Hg(II)_i$ species which are more readily taken up by bacteria via passive diffusion.[49-51] In this study, we observed no correlation between the predicted concentration of neutral $Hg(II)_i$ species and cell-normalized MeHg production (Supporting Information **Figure SI-2a**). Cell-normalized MeHg production was weakly negatively correlated (r^2 = 0.19, p = 0.06) with predicted total dissolved Hg concentration, suggesting that differences in metacinnabar solubility were not driving observed differences in MeHg production. As shown in **Figure 3**, a positive correlation is observed (r^2 = 0.54, p < 0.001), however, between the log of the predicted $Hg(SR)_2$ concentration and the log of cell-normalized MeHg production across all experiments. Two points are noteworthy about this observation. First, much (54%) of the variation in cell-normalized Hg-methylation can be explained by the predicted $Hg(SR)_2$ concentration despite significant differences in DOM aromaticity, ^{201}THg concentration, cell density, and generally smaller differences in sulfide, pH, and DOM concentration across experiments. Second, data for Hg methylation by ND132 in the presence of 500 µM L-cysteine

fall roughly along the regression line (slope of regression line decreases from 0.25 to 0.17 when including the cysteine data, r^2 increases to 0.64). Schaefer et al. observed similar rates of Hg methylation by ND132 in the presence of three thiol containing amino acids/peptides (cysteine, penicillamine, and glutathione).[36] And recently, Mazrui *et al.*[12], in sediment microcosm experiments, demonstrated enhanced Hg methylation when sediments were amended with Hg-DOM complexes under conditions of undersaturation with respect to metacinnabar. These workers hypothesized either direct uptake of Hg-DOM complexes (presumably Hg-DOM thiol complexes) or DOM ligands acting as a shuttle between solution and the cell surface metal ion transporters hypothesized[48] to be involved in $Hg(II)_i$ uptake. Integrating these observations together with the data presented here, we can hypothesize that complexes of $Hg(II)_i$ with DOM thiols have similar bioavailability as $Hg(II)_i$ complexes with low molecular weight thiols such as cysteine. As noted above, Hg-DOM thiol complexes are predicted to be a small (<0.01 to 2.4%) fraction of the total Hg pool in these experiments, and predicted $Hg(SR)_2$ concentrations are in many cases several orders of magnitude smaller than observed MeHg concentrations (10^{-15}—10^{-12} M for $Hg(SR)_2$ vs. 10^{-11}—10^{-10} M for MeHg). The concentration of available DOM thiols is predicted to be in large excess of $Hg(II)_i$ (see Supporting Information Table SI-4), however, allowing for continual $Hg(SR)_2$ complex formation with intracellular compartmentalization of $Hg(II)_i$ and subsequent methylation. The observed MeHg production could be driven by a small pool of highly bioavailable $Hg(SR)_2$ complexes if formation of new $Hg(SR)_2$ complexes to replace $Hg(SR)_2$ complexes taken up by cells is sufficiently rapid. Hg methylation efficiencies less than 100% may reflect competition between Hg internalization and methylation and sorption to the cell wall/membrane or other

412 cellular components.[52] Several studies suggest that equilibrium of $Hg(II)_i$ with strong S-donor
413 ligands occurs on timescales of hours-days,[26,53] raising the possibility that rates of ligand-
414 exchange may play a role in microbial Hg methylation.[37,52]

415 To summarize, sulfurization of DOM clearly increases the bioavailability of $Hg(II)_i$ in
416 Hg-sulfide-DOM solutions. Two plausible, non-mutually exclusive, mechanisms for this
417 increase in bioavailability are that DOM thiols stabilize HgS clusters/nanoparticles highly
418 bioavailable for uptake and/or that DOM thiols form complexes with $Hg(II)_i$ directly taken up
419 or readily exchanged on the cell surface. At present, we cannot clearly distinguish between
420 these two mechanisms, in part, because the low Hg concentrations necessary to mimic
421 environmental conditions (and realistic Hg/DOM ratios) are inaccessible to direct
422 spectroscopic investigation of $Hg(II)_i$ speciation. In either case, DOM sulfurization profoundly
423 alters Hg uptake and methylation.

424 *3.4 Environmental Implications*

425 As demonstrated in this paper, DOM sulfurization can play an important role in the
426 biogeochemical cycling of chalcophilic trace elements such as Hg. Increased DOM
427 sulfurization leads to enhanced microbial production of the potent neurotoxin MeHg under
428 conditions where Hg-DOM thiol complexes are predicted to be only minor $Hg(II)_i$ species.
429 While there have been significant advances in total thiol quantification in soil and sediment
430 porewaters,[54] investigations of DOM composition using FT-ICR MS[18-20] suggest that both the
431 molecular diversity and concentration of DOM thiols in soils and sediment porewaters may be
432 greater than previously realized. For example, in the highly eutrophied portions of the
433 Northern Florida Everglades, where porewater sulfide concentrations can reach the several

hundred μM level and DOC concentrations range from 40-100 mg/L, porewater DOM thiols produced by sulfurization likely reach upwards of 30 μM.[20] Interestingly, we also observed that organic matter more "terrestrial" in nature (higher aromaticity and average molecular weight) has greater capacity for sulfurization. While natural or anthropogenic sulfate enrichment of terrestrial ecosystems undoubtedly contributes to stimulation of sulfate-reducing bacteria, an important group of Hg methylators,[55] sulfurization of DOM may also enhance $Hg(II)_i$ bioavailability for methylation in such environments.

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618 **Table 1.** Select characteristics of dissolved organic matter isolates utilized in this study. Superscripts indicate reference for
619 characterization data.

		Sulfur content and speciation					Carbon distribution by ^{13}C NMR[a]			
Isolate	**Description**	**Native S content (weight %)[a]**	**Native S/C ratio (mmol S/mol C) of raw DOM[a]**	**Measured S/C ratio of unsulfurized SPE DOM (mmol S/mol C)[b]**	**%S exocyclic[c]**	**%S hetero-cyclic[c]**	**% aromatic**	**% carbonyl**	**% carboxyl**	**% acetal and (hetero) aliphatic**
Suwannee River humic acid (SRHA, standard II)	Blackwater river draining Okeefenokee Swamp, sampled at Fargo GA	0.54	4.14	3.42±0.42	23.6	40.4	31	6	15	49
Suwannee River fulvic acid (SRFA, standard II)	Blackwater river draining Okeefenokee Swamp, sampled at Fargo GA	0.46	3.15	1.88±0.23	25.8	27.6	22	5	17	57
Nordic Lake fulvic acid	Drinking water reservoir in Vallsjøen, Skarnes, Norway	0.58	3.30	3.12±0.39	n.d.	n.d.	31	10	24	37
Pony Lake fulvic acid	Shallow eutrophic saline pond in Anarctica	3.03	21.7	11.0±1.4	46.9	21.3	12	1.2	17	69.6

620
621
622 [a]Data from International Humic Substances Society. Note that percentages for carbon distribution do not necessarily add to 100.
623 Other elemental composition data for these samples also available at http://www.humicsubstances.org/elements.html.
624 [b]This study. Measured S/C ratio on native (unsulfurized) DOM sample after recovery by solid phase extraction (SPE).
625 [c]Data from Manceau and Nagy[21] determined by XANES spectroscopy.
626 n.d. = no data available

Figure Captions

Figure 1. a) Extent of total sulfur incorporation into dissolved organic matter as a function of the added sulfide/DOC ratio. **b)** Relationship between bulk DOM functional group content determined by ^{13}C NMR and the total sulfur incorporation capacity. Sulfur incorporation capacity defined as the maximum change in DOM S content upon reaction with sulfide. For the NLFA sample and the Everglades DOM[20], a plateau in DOM S content was not observed, so the reported S incorporation capacity should be interpreted as a minimum value. PLFA = Pony Lake fulvic acid; SRHA = Suwannee River humic acid; SRFA = Suwannee River fulvic acid; NLFA = Nordic lake fulvic acid. ESHA = Elliott soil humic acid. Data for ESHA from Hoffmann *et al.*[16] Data for the Everglades DOM from Poulin *et al.*,[20] with functional group content estimated based on data in Waples *et al.*[33]

Figure 2. Methylmercury production by *Desulfovibrio desulfuricans* ND132 in Hg methylation assays as a function of the S/C ratio of dissolved organic matter. Methylation assays were conducted with 4.5±1.6 x 10^8 cells/mL at pH 7.36±0.15 and contained 8.3 to 10 mg C/L of DOM, and 3.4±2.0 µM total sulfide (full details of experimental conditions in Supporting Information Table SI-3). **a)** Total enriched ^{201}Hg in medium; **b)** absolute production of enriched $Me^{201}Hg$; **c)** fraction of total ^{201}Hg methylated; **d)** cell-normalized $Me^{201}Hg$ production. Shown in panels c and d are mean results for DOM-free controls and experiments with 500 µM L-cysteine for reference. Error bars are standard deviations of triplicate experiments, excepting NLFA experiments that were performed in duplicate.

Figure 3. Relationship between the predicted concentration of $Hg(SR)_2$ complexes, where SR is an organic thiol (either DOM thiol or L-cysteine), and the cell-normalized MeHg production. Predicted $Hg(SR)_2$ concentrations at equilibrium were calculated based on experimental pH, estimated organic thiol concentration, total Hg concentration, and sulfide concentrations (see text and Supporting Information for details).

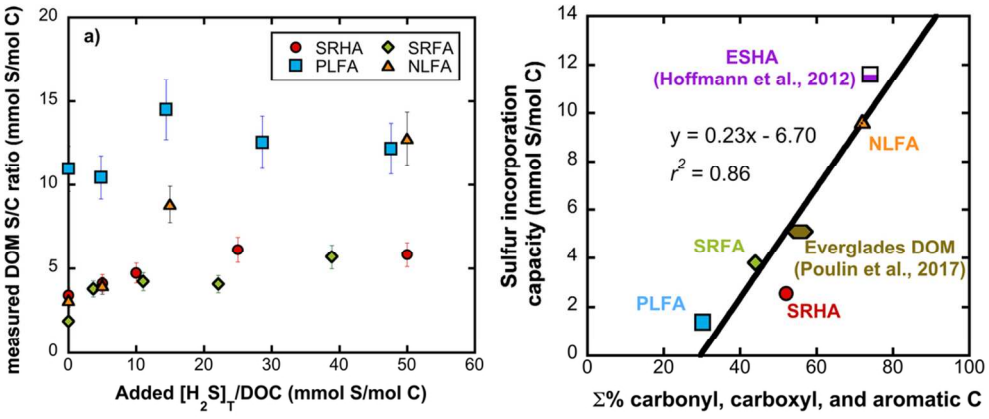


Figure 1

50x21mm (600 x 600 DPI)

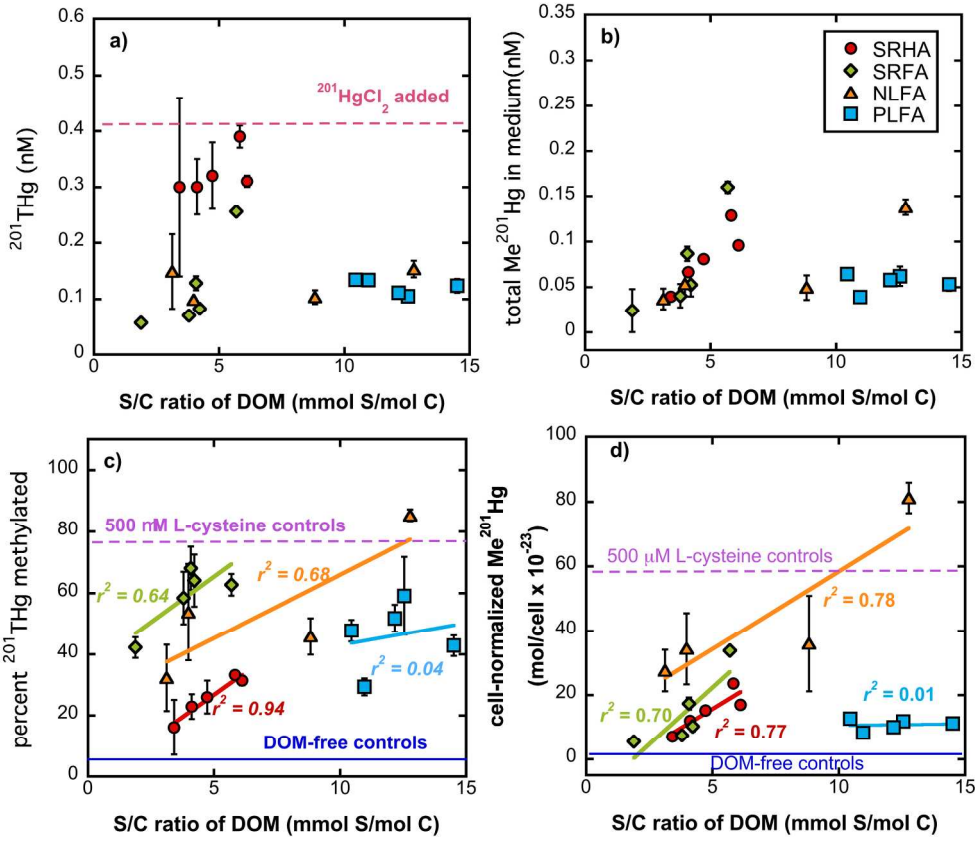


Figure 2

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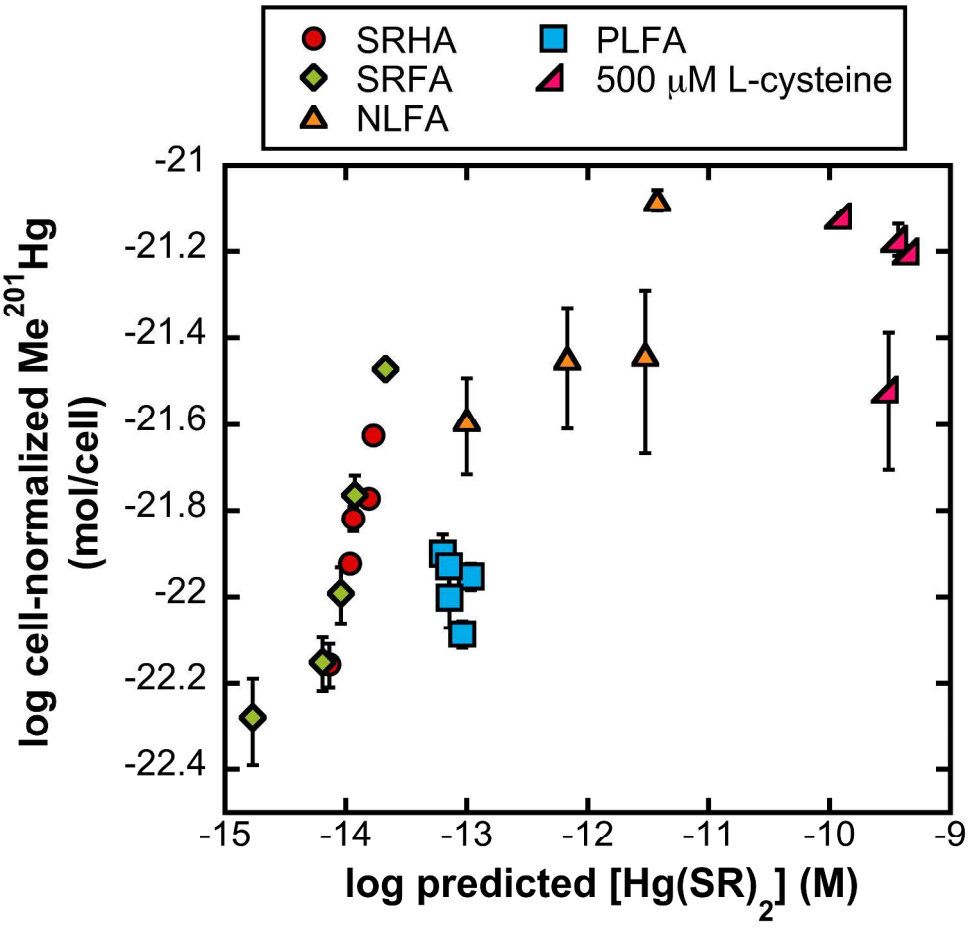


Figure 3

165x155mm (600 x 600 DPI)

DOM sulfurization

Inhibition of metacinnabar (β-HgS) growth/aggregation and $Hg(SR)_2$ complex formation

Enhanced microbial Hg methylation

